Two-dimensional position sensing using absolute-type encoder scheme employing pseudorandom binary arrays

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A novel absolute-type encoder scheme employing pseudorandom binary arrays for two-dimensional position sensing is described. With the encoder scheme presented, combined measurement of shaft angular and translational position using a single transducer is possible. Brief details on design, construction and testing are given.

Introduction: The accurate measurement of position is of great importance in many industrial applications [1]. Angular position, if required, can be measured by means of an optical shaft encoder that is either incremental or absolute; and when linear or translational motion is to be measured, a rotary encoder can be attached to a mechanical device such as a lead screw, or alternatively a linear optical encoder can be employed. In applications such as robotics where an actuator, or combination of actuators, results in motion in two dimensions and combinations of both angular and translational motion are common, a number of encoders must be used and their outputs combined to give accurate positional information. This requirement increases the cost and complexity of the instrumentation required. Consequently, potential advantages can result if two-dimensional positional information is recovered using a single sensing device.

In recent years, a number of possible coding and scanning methods for two-dimensional position sensing have been reported in the literature [2 – 4] In [4], for example, use of pseudorandom binary arrays (PRBAs) [5] for two-dimensional position recovery for a free-ranging automated guided vehicle is described. The purpose of this Letter is to present, with the aid of an example, a further application of PRBAs for two-dimensional position sensing. Namely, a novel encoder scheme of the absolute type for the combined measurement of actuator shaft angular and translational position using a single transducer is described.

Table 1 Sample available PRBAs and primitive polynomials $h(x)$ up to $N = 65535$ suitable for transducer realisation

n	$N = 2^n-1$	$h(x)$	Sample PRBAs ($n_1 \times n_2$)	Sensor array ($k1 \times k2$)
4	15	x^4+x+1	(3×5)	(2×2)
6	63	x^6+x+1	(7×9)	(3×2)
8	255	$x^8+x^6+x^5+x+1$	(3×85)(15×17)	(2×4)(4×2)
9	511	x^9+x^4+1	(7×73)	(3×3)
10	1023	$x^{10}+x^3+1$	(3×341)(31×33)	(2×5)(5×2)
12	4095	$x^{12}+x^7+x^4+x^3+1$	(7×585)(63×65)	(3×4)(6×2)
14	16383	$x^{14}+x^{12}+x^{11}+x+1$	(3×5461)(127×129)	(2×7)(7×2)
15	32767	$x^{15}+x+1$	(7×4681)(31×1057)	(3×5)(5×3)
16	65535	$x^{16}+x^5+x^3+x^2+1$	(3×21845)(15×4369) (255×257)	(2×8)(4×4) (8×2)

PRBAs and two-dimensional position sensing: PRBAs can be constructed from n-stage binary shift registers generating pseudorandom binary sequences $a_1, a_2, ..., a_N$ of length $N = 2^n - 1$. PRBAs exist whenever $N = 2^n - 1$ can be factored into two or more relatively prime factors. In one approach for PRBA construction, the sequence folding method [6], the pseudorandom sequence used is plotted as an array of n_1 rows and n_2 columns. The plotting process starts at the top left-hand corner of the array and proceeds down the main diagonal. When an edge of the array is reached, the array is re-entered on the opposite edge using the next row or column. Provided n_1 and n_2 are relatively prime, one pass through the plotting process with the selected sequence will result in either a 0 or 1 being placed only once in each cell of the array, and the process will end at the bottom right-hand corner. If an appropriately sized $k1$-by-$k2$ window is placed over a PRBA, each of the $2^{k1k2} - 1$ nonzero arrays that result is unique, and is therefore seen only once. This is known as the window property of PRBAs. For the generated PRBAs discussed here, the following applies [6]: $2^n - 1 = 2^{k1k2} - 1$ such that $n_1 = 2^{k1} - 1$ and $n_2 = (2^n - 1)/n_1$ are relative prime and greater than 1. Then $2^n - 1 = n_1 n_2$.

A tabulation of available PRBAs over $GF(2)$ for $N = 2^n - 1$ and $N = n_1 \times n_2$ up to $N = 65535$ is given in [5], and primitive polynomials $h(x)$ for the n-stage binary shift registers required to generate pseudorandom binary sequences needed for PRBA construction are listed in [6]. Using this information, Table 1, which lists sample available PRBAs, can be obtained. A first step in the realisation of a transducer is to select an appropriately sized array from Table 1. Next, from the Table, a primitive polynomial and corresponding binary shift register with suitable feedback are obtained and the detailed two-dimensional pattern of the chosen PRBA revealed.

For two-dimensional position sensing, the window property of PRBAs is invoked. In practice, to employ this property, the chosen PRBA is laid down as a pattern on the sensed surface, and a sensor head consisting of a $k1$-by-$k2$ array of individual sensors is used to detect, at its current position, which one of the $2^{k1k2} - 1$ possible nonzero binary $k1$-by-$k2$ arrays is present. In practice, to achieve this, either optical, magnetic or capacitive sensing methods can be used. Since the relationship between these array windows and their two-dimensional position is known, a look-up table approach can be used to recover the required two-dimensional positional information.

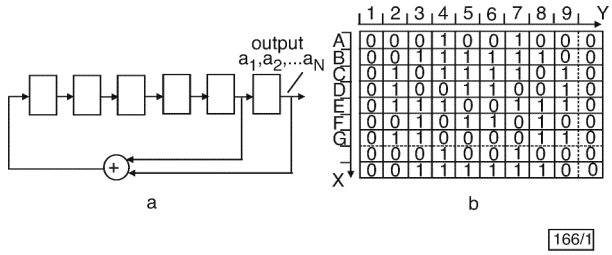


Fig. 1 *Shift register and generated PRBA*

a Shift register for primitive polynomial $h(x) = x^6 + x + 1$ used to construct PRBA
b Generated PRBA with two example 3-by-2 windows

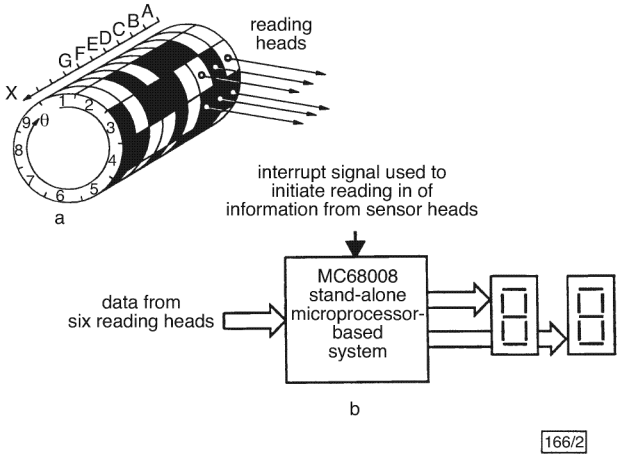


Fig. 2 *Recovery arrangement and stand-alone microprocessor-based system*

a Arrangement for recovery of shaft translational position, X, and angular position, θ
b Stand-alone microprocessor-based system used to implement encoder scheme

Example of absolute-type position transducer: The starting point in the realisation of a transducer is to select from Table 1 an appropriately sized PRBA and a generating pseudorandom sequence. For the example presented here, a 7-by-9 PRBA is used with a shift register configuration, implementing primitive polynomial $h(x) = x^6 + x + 1$, as shown in Fig. 1*a*. Using the sequence folding method, and taking the initial shift register state as the binary state 100000, results in the PRBA given in Fig. 1*b*. Also shown in this Figure are two sample 3-by-2 windows. For this PRBA: $N = 63 = 2^6 - 1$ with $k1 = 3$, $k2 = 2$, $n_1 = 7$ and $n_2 = 9$. That is, a 7 row by 9 column array is generated with 63 unique 3-by-2 windows. Consequently, the position of a six-sensor reading head can be recovered in the two-dimensional X, Y co-ordinate system shown. Using a look-up table method, a sensor head placed at, for example, window position [01, 11, 01] can be identified as occupying location A,3, and when placed at array position [00, 10, 11] location G,4 is returned, taking the top left-hand corner of each window as a reference point. Note that to accommodate all 63 unique 3-by-2 windows the top two rows have to be repeated at the bottom of the array and the first column repeated after the last column in the array.